FP_TechnicalDoc_MaryAnnMastri_Individual_ITAI2373

NewsBot 2.0 — Technical Documentation

Final Project — ITAI 2373
Student: Mary Ann Mastri
Group: Individual
Date: May 2, 2025

Table of Contents

1. Introduction
2. System Architecture
3. Core NLP Components
4. Multilingual Intelligence Layer
5. Conversational Interface
6. Content Enhancement Engine
7. Semantic Search Engine
8. Training Process
9. Testing & Evaluation
10. Final Demo Overview
11. Conclusion

1. Introduction

NewsBot 2.0 is a production-ready, multilingual news analysis platform designed to process, interpret, and summarize news content using advanced natural language processing (NLP) techniques. This system expands the midterm NewsBot into a comprehensive intelligence engine capable of classification, topic modeling, sentiment analysis, summarization, semantic search, multilingual processing, and conversational interaction.

The goal of this project is to demonstrate mastery of modern NLP methods, modular system design, and professional-grade software engineering practices.

2. System Architecture

NewsBot 2.0 is built using a layered, modular architecture:

2.1 Architecture Overview

Input → Intent Classification → Analysis Pipeline → Output

2.2 Major Subsystems

- Core NLP Engine
- Multilingual Processing Layer

- Conversational Interface
- Semantic Search Engine
- Content Enhancement Engine
- Evaluation Suite

This architecture ensures scalability, maintainability, and clear separation of concerns.

3. Core NLP Components

3.1 AdvancedNewsClassifier

- TF-IDF vectorization
- Logistic regression classifier
- Predicts news category with confidence score

3.2 TopicDiscoveryEngine

- Gensim LDA topic model
- Dictionary + corpus creation
- Topic inference for new articles

3.3 SentimentEvolutionTracker

- VADER sentiment analyzer
- Returns compound sentiment score

3.4 EntityRelationshipMapper

- spaCy NER
- Subject-Verb-Object extraction
- Lightweight relationship mapping

3.5 IntelligentSummarizer

- Extractive summarization
- Summary quality scoring

4. Multilingual Intelligence Layer

4.1 MultilingualProcessor

- Language detection
- Translation to English
- Translation back to original language

4.2 MultilingualNER

- Wraps EntityRelationshipMapper
- Ensures NER works across languages
- Maintains consistent output structure

5. Conversational Interface

The ConversationalInterface enables natural language interaction.

5.1 IntentClassifier

- Rule-based keyword mapping
- Routes queries to correct subsystem

5.2 ConversationalInterface

- Processes user queries
- Handles category, topics, sentiment, summary, insights
- Includes full article analysis wrapper

6. Content Enhancement Engine

ContentEnhancer

- Generates insights across multiple articles
- Identifies topic clusters
- Produces high-level summaries

7. Semantic Search Engine

SemanticSearchEngine

- Embedding generation using SentenceTransformer
- Cosine similarity search
- Supports “find similar articles” queries

8. Training Process

8.1 Dataset Loading

- Extracted from ZIP
- Loaded into DataFrames
- Cleaned and preprocessed

8.2 Model Training

- Classifier trained on labeled categories

- Topic model trained on article corpus
- Semantic search engine encodes documents

9. Testing & Evaluation

9.1 Component Tests

- Classification
- Topic modeling
- Sentiment
- NER
- Summarization
- Semantic search
- Multilingual detection

9.2 Integration Test

Ensures analyze_article() returns:

- classification
- topics
- sentiment
- entities
- relationships
- summary

9.3 Performance Test

- Measures average processing time
- Ensures system responsiveness

9.4 Edge Case Tests

- Empty text
- Non-English text
- Very short text
- Extremely long text

All tests passed successfully.

10. Final Demo Overview

The final demo showcases:

- Full article analysis
- Conversational queries

- Semantic search
- Insight generation
- Multilingual processing

11. Conclusion

NewsBot 2.0 is a robust, modular, multilingual news intelligence system demonstrating advanced NLP mastery and production-ready architecture. It fulfills all requirements of the final project and provides a strong foundation for real-world applications.